

Integrated M.S.–Ph.D. Student, [IPA Laboratory](#),
Department of Electrical Engineering,
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Research Profile

Diffusion Models and Reliable Generation

Reference-based diffusion for image super-resolution and restoration (ICCV 2025, ECCV 2026), and reliable sampling guidance for hallucination-resistant generation (TAG, ICML 2026), toward high-fidelity content restoration and enhancement.

Associative Memory and Efficient Visual Representations

Hopfield-perspective analysis of attention for balancing fidelity and diversity in generative models (ICML 2026); efficient implicit neural representations for visual data (ECCV 2026, CVPR 2025).

Education

Korea University / 高麗大學校

Integrated M.S.–Ph.D. in Electrical Engineering
Advisor: Kyong Hwan Jin

Republic of Korea

MAR. 2024 - PRESENT, GPA: 4.31/4.5

Gachon University / 嘉泉大學校

B.S. in Software (Computer Science)

Republic of Korea

MAR. 2020 - FEB. 2024, GPA: 4.29/4.5

Selected Publications

♥: Equal contribution, †: Corresponding author

1. **Hyunmin Cho**, Jaejun Yoo and Kyong Hwan Jin[†], “Recurrent Sinusoidal INRs for Efficient High-Fidelity Representation”, The 19th European Conference on Computer Vision (**ECCV**), 2026. [\[PDF\]](#) [\[Project\]](#)
2. **Hyunmin Cho**, Woo Kyoung Han and Kyong Hwan Jin[†], “Balancing Fidelity and Diversity in Diffusion Models via Symmetric Attention Decomposition: Hopfield Perspective”, Forty-Third International Conference on Machine Learning (**ICML**), 2026. [\[PDF\]](#) [\[Project\]](#)
3. **Hyunmin Cho**♥, Donghoon Ahn♥, Susung Hong♥, Jee Eun Kim, Seungryong Kim[†] and Kyong Hwan Jin[†], “TAG: Tangential Amplifying Guidance for Hallucination-Resistant Sampling”, Forty-Third International Conference on Machine Learning (**ICML**), 2026. [\[PDF\]](#) [\[Project\]](#)
4. Byeonghun Lee♥, **Hyunmin Cho**♥, Honggyu Choi, Soo Min Kang, Iljun Ahn and Kyong Hwan Jin[†], “Reference-based Super-Resolution via Image-based Retrieval-Augmented Generation Diffusion”, IEEE/CVF International Conference on Computer Vision (**ICCV**), 2025. [\[PDF\]](#) [\[Project\]](#)
5. Byeonghun Lee, **Hyunmin Cho**, Sunghoon Im[†] and Kyong Hwan Jin[†], “ReAL: Reference-to-Image (R2I) Aware Latent Diffusion for Image Super-Resolution”, The 19th European Conference on Computer Vision (**ECCV**), 2026.
6. Woo Kyoung Han, Byeonghun Lee, **Hyunmin Cho**, Sunghoon Im[†] and Kyong Hwan Jin[†], “Towards Lossless Implicit Neural Representation via Bit Plane Decomposition”, IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2025. [\[PDF\]](#) [\[Project\]](#)

Research Experience

Samsung Research

JAN. 2026 – DEC. 2026

Development of Hallucination Mitigation Techniques for Generative Models

- Leading the development of an uncertainty-driven world-model framework for hallucination-resistant video reasoning, using structured scene graphs and first-order logic to verify and correct uncertain VLM predictions.

Samsung Research

JUN. 2024 – MAY 2025

Development of User-Preferred Image-Based Image Quality Technology

- Co-developed iRAG, a retrieval-augmented diffusion framework with reference retrieval and hallucination filtering for realistic image super-resolution (ICCV 2025).

Awards

Gold Prize, 38th Workshop of Image Processing and Image Understanding (IPIU), [\[Cert.\]](#)

2026

Academic Service

Reviewer, NeurIPS 2026

Reference

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